ST2334
AY25/26 Sem 2
by ngmh

Basic Concepts of Probability

Union
- $A \cup B = \{x : x \in A \lor x \in B\}$
- $\bigcup_{i=1}^{n} A_i = A_1 \cup A_2 \cup ... \cup A_n = \{x : x \in A_1 \lor x \in A_2 \lor ... \lor x \in A_n\}$

Intersection
- $A \cap B = \{x : x \in A \land x \in B\}$
- $\bigcap_{i=1}^{n} A_i = A_1 \cap A_2 \cap ... \cap A_n = \{x : x \in A_1 \land x \in A_2 \land ... \land x \in A_n\}$

Complement $A' = \{x : x \in S \land x \notin A\}$
Mutually Exclusive / Disjoint $A \cap B = \emptyset$
Event Operations
- $A \cap A' = \emptyset$
- $A \cap \emptyset = \emptyset$
- $A \cup A' = S$
- $(A')' = A$
- $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
- $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- $A \cup B = A \cup (B \cap A')$
- $A = (A \cap B) \cup (A \cap B')$

De Morgan's Law
- $(A_1 \cup A_2 \cup ... \cup A_n)' = A_1' \cap A_2' \cap ... \cap A_n'$
- $(A_1 \cap A_2 \cap ... \cap A_n)' = A_1' \cup A_2' \cup ... \cup A_n'$

Permutation Selection and arrangement of r objects out of n: $P_r^n = \frac{n!}{(n-r)!}$
Combination Selection of r objects out of n without regards to order: $C_r^n = \binom{n}{r} = \frac{n!}{r!(n-r)!}$, $\binom{n}{r} = \binom{n}{n-r}$
Probability Axioms
1. $0 \leq P(A) \leq 1$
2. $P(S) = 1$
3. $A \cap B = \emptyset \Rightarrow P(A \cup B) = P(A) + P(B)$

Basic Properties of Probability
1. $P(\emptyset) = 0$
2. For mutually exclusive events, $P(A_1 \cup A_2 \cup ... \cup A_n) = P(A_1) + P(A_2) + ... + P(A_n)$
3. $P(A') = 1 - P(A)$
4. $P(A) = P(A \cap B) + P(A \cap B')$
5. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
6. If $A \subset B$ then $P(A) \leq P(B)$

Conditional Probability The probability of B given A occurring is $P(B|A) = \frac{P(A \cap B)}{P(A)}$
Multiplication Rule $P(A \cap B) = P(A)P(B|A) = P(B)P(A|B)$ given $P(A) \neq 0$ or $P(B) \neq 0$
Inverse Probability Formula $P(A|B) = \frac{P(A)P(B|A)}{P(B)}$
Independence $P(A \cap B) = P(A)P(B)$, denoted $A \perp B$
Some Properties of Independence
- If $P(A) > 0, P(B) > 0, A \perp B$, then A and B are not mutually exclusive
- If $P(A) > 0, P(B) > 0$, and they are mutually exclusive, then $A \not\perp B$
- S and $\emptyset$ are independent of any other event
- If $A \perp B$, then $A \perp B', A' \perp B, A' \perp B'$

Partition For mutually exclusive events with $\bigcup_{i=1}^{n} A_i = S$, $A_1, A_2, ..., A_n$ is a partition of S
Law of Total Probability For a partition of S, $P(B) = \sum_{i=1}^{n} P(B \cap A_i) = \sum_{i=1}^{n} P(A_i)P(B|A_i)$
Baye's Theorem For a partition of S, $P(A_k|B) = \frac{P(A_k)P(B|A_k)}{\sum_{i=1}^{n} P(A_i)P(B|A_k)}$

Random Variables

Probability Mass Function for Discrete Random Variables

$$f(x) = \begin{cases} P(X = x) & x \in R_X \\ 0 & x \notin R_X \end{cases}$$

1. $f(x_i) \geq 0$ for all $x_i \in R_X$
2. $f(x) = 0$ for all $x \notin R_X$
3. $\sum_{i=1}^{\infty} f(x_i) = 1$

Probability Density Function for Continuous Random Variables
1. $f(x) \geq 0$ for all $x \in R_X$, $f(x) = 0$ for all $x \notin R_X$
2. $\int_{R_X} f(x)dx = 1$
3. For $a \leq b$, $P(a \leq X \leq b) = \int_a^b f(x)dx$

It suffices to check conditions 1 and 2
Cumulative Distribution Function $F(x) = P(X \leq x)$
- Discrete RV: $F(x) = \sum_{t \in R_X, t \leq x} P(x = t)$, $P(a \leq X \leq b) = F(b) - F(a-)$
- Continuous RV: $F(x) = \int_{-\infty}^{x} f(t)dt$, $f(x) = \frac{dF(x)}{dx}$, $P(a \leq X \leq b) = F(b) - F(a)$

Expectation / Mean
- Discrete RV: $\mu_X = E(X) = \sum_{x_i \in R_X} x_i f(x_i)$
- Continuous RV: $\mu_X = E(X) = \int_{-\infty}^{\infty} xf(x)dx = \int_{x \in R_X} xf(x)dx$

Properties of Expectation
1. $E(aX + b) = aE(X) + b$
2. $E(X + Y) = E(X) + E(Y)$
3. For arbitrary $g(.)$
   - Discrete RV: $E[g(X)] = \sum_{x \in R_X} g(x)f(x)$
   - Continuous RV: $E[g(X)] = \int_{R_X} g(x)f(x)dx$

Variance and Properties $\sigma_X^2 = V(X) = E(X - \mu_X)^2$
- Discrete RV: $V(X) = \sum_{x \in R_X} (x - \mu_X)^2 f(x)$
- Continuous RV: $V(X) = \int_{-\infty}^{\infty} (x - \mu_X)^2 f(x)dx$
- $V(X) \geq 0$ for any X, with equality holding iff $P(X = E(X)) = 1$
- $V(aX + b) = a^2 V(X)$
- Alternative Formula: $V(X) = E(X^2) - [E(X)]^2$

Standard Deviation $\sigma_X = \sqrt{V(X)}$, $\sigma_X \geq 0$

Joint Distributions

Discrete Joint Probability Function $f_{X,Y}(x, y) = P(X = x, Y = y)$, $(x, y) \in R_{x,y}$
1. $f_{X,Y}(x, y) \geq 0$, $(x, y) \in R_{X,Y}$
2. $f_{X,Y}(x, y) = 0$, $(x, y) \notin R_{X,Y}$
3. $\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} f_{X,Y}(x_i, y_j) = \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} P(X = x_i, Y = y_j) = 1$ or $\sum\sum_{(x,y) \in R_{X,Y}} f(x, y) = 1$
4. $P((X, Y) \in A) = \sum\sum_{(x,y) \in A} f(x, y)$

Continuous Joint Probability Function
$P((X, Y) \in D) = \int\int_{(x,y) \in D} f_{X,Y}(x, y)dydx$, or
$P(a \leq X \leq b, c \leq Y \leq d) = \int_a^b \int_c^d f_{X,Y}(x, y)dydx$
1. $f_{X,Y}(x, y) \geq 0$, $(x, y) \in R_{X,Y}$
2. $f_{X,Y}(x, y) = 0$, $(x, y) \notin R_{X,Y}$
3. $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y)dxdy = 1$ or $\int\int_{(x,y) \in R_{X,Y}} f_{X,Y}(x, y)dxdy = 1$

Marginal Probability Distribution Projection of 2D function onto 1D function
- Y Discrete: $f_X(x) = \sum_y f X, Y(x, y)$
- Y Continuous: $f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y)dy$

Conditional Distribution Distribution of Y given that X is observed, $f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)}$
1. Defined for $f_X(x) > 0$
2. If $f_X(x) > 0$, then $f_{X,Y}(x, y) = f_X(x)f_{Y|X}(y|x)$
3. $P(Y \leq y|X = x) = \int_{-\infty}^{y} f_{Y|X}(y|x)dy$
4. $E(Y|X = x) = \int_{-\infty}^{\infty} yf_{Y|X}(y|x)dy$

Independent Random Variables $X_1, X_2, ..., X_n$ are independent iff $f_{X_1,X_2,...,X_n}(x_1, x_2, ..., x_n) = f_{X_1}(x_1)f_{X_2}(x_2)...f_{X_n}(x_n)$. $R_{X,Y}$ must be a product space of $R_X \times R_Y$ by satisfying all possibilities.
Properties of Independent Random Variables
1. $P(X \in A; Y \in B) = P(X \in A)P(Y \in B)$, for any real x, y, $P(X \leq x; Y \leq y) = P(X \leq x)P(Y \leq y)$
2. For arbitrary $g_1(.)$ and $g_2(.)$, $g_1(X)$ and $g_2(Y)$ are independent
3. If $f_X(x) > 0$, then $f_{Y|X}(y|x) = f_Y(y)$

Checking Independence
1. $R_{X,Y}$ where the probability function is positive must be a product space
2. For $(x, y) \in R_{X,Y}$, $f_{X,Y}(x, y) = C \times g_1(x) \times g_2(y)$, i.e. it can be factorised into terms independent of other variables

Marginal Distribution under Independence Given $g_1(x)$ and $g_2(y)$,
- X Discrete: Probability mass function is $f_X(x) = \frac{g_1(x)}{\sum_{t \in R_X} g_1(t)}$
- X Continuous: Probability density function is $f_X(x) = \frac{g_1(x)}{\int_{t \in R_X} g_1(t)dt}$

Expectation
- Discrete: $E(g(X, Y)) = \sum_x \sum_y g(x, y)f_{X,Y}(x, y)$
- Continuous: $E(g(X, Y)) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y)f_{X,Y}(x, y)dydx$

Covariance $cov(X, Y) = E[(X - E(X))(Y - E(Y))]$ using $g(X, Y) = (X - E(X))(Y - E(Y))$
- Discrete: $cov(X, Y) = \sum_x \sum_y (x - \mu_X)(y - \mu_Y)f_{X,Y}(x, y)$
- Continuous: $cov(X, Y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \mu_X)(y - \mu_Y)f_{X,Y}(x, y)dxdy$

Properties of Covariance
1. $cov(X, Y) = E(XY) - E(X)E(Y)$
2. If X and Y are independent, $cov(X, Y) = 0$, and $E(XY) = E(X)E(Y)$
3. $cov(aX + b, cY + d) = ac \cdot cov(X, Y)$ using $cov(X + b, Y) = cov(X, Y)$ and $cov(aX, Y) = acov(X, Y)$
4. $V(aX + bY) = a^2V(X) + b^2V(Y) + 2ab \cdot cov(X, Y)$ using $V(X + Y) = V(X) + V(Y) + 2cov(X, Y)$

Properties of Variance and Covariance
1. $V(X_1 + X_2 + ... + X_n) = V(X_1) + V(X_2) + ... + V(X_n) + 2\sum_{j>i} cov(X_i, X_j)$
2. For independent $X_1, X_2, ..., X_n$, $V(X_1 \pm X_2 \pm ... \pm X_n) = V(X_1) + V(X_2) + ... + V(X_n)$

Correlation Coefficient $\rho_{X,Y} = \frac{cov(X,Y)}{\sigma_X \sigma_Y}$

Special Probability Distributions

Discrete Distributions

Discrete Uniform Distribution The random variable assumes discrete values with equal probability
- pmf: $1/k$ if there are k values it can take on
- $\mu_X = E(X) = \frac{1}{k}\sum_{i=1}^{k} x_i$
- $\sigma_X^2 = V(X) = E(X^2) - [E(X)]^2 = \frac{1}{k}\sum_{i=1}^{k} x_i^2 - \mu_X^2$

Bernoulli Trial Random experiment with only two possible outcomes
Bernoulli Random Variable $X \sim Bernoulli(p)$ Success of a Bernoulli trial
- pmf: $p, x = 1; 1 - p = q, x = 0$
- $f_X(x) = p^x(1 - p)^{1-x}, x = 0, 1$
- $\mu_X = E(X) = p$
- $\sigma_X^2 = V(X) = p(1 - p) = pq$

Bernoulli Process Sequence of repeatedly performed independent and identical Bernoulli trials
Binomial Random Variable $X \sim Bin(n, p)$ Counts number of successes in n trials of a Bernoulli Process

- $P(X=x)=\binom{n}{x}p^x(1-p)^{n-x}, 0\le x\le n$
- $E(X)=np$
- $V(X)=np(1-p)$

Negative Binomial Distribution $X\sim NB(k,p)$ Let X be the number of independent and identically distributed Bernoulli trials needed until the kth success occurs. Then X follows a Negative Binomial Distribution.
- pmf: $f_X(x)=P(X=x)=\binom{x-1}{k-1}p^k(1-p)^{x-k}, x\ge k$
- Derived from $k-1$ success in first $x-1$ trials and success on xth trial
- $E(X)=\frac{k}{p}$
- $V(X)=\frac{(1-p)k}{p^2}$

Geometric Distribution $X\sim Geom(p)$ Let X be the number of independent and identically distributed Bernoulli trials needed until the first success occurs. Then X follows a geometric distribution.
- pmf: $f_X(x)=P(X=x)=(1-p)^{x-1}p$
- $E(X)=\frac{1}{p}$
- $V(X)=\frac{1-p}{p^2}$

Hypergeometric Distribution $X\sim HG(N,M,n)$ Suppose a sample of size n is drawn without replacement from a finite population of size N containing M successes and $N-M$ failures. Let X be the number of successes in the sample. Then X follows a hypergeometric distribution.
- pmf: $f_X(x)=P(X=x)=\frac{\binom{M}{x}\binom{N-M}{n-x}}{\binom{N}{n}}, \max(0,n-(N-M))\le x\le \min(n,M)$
- $E(X)=n\frac{M}{N}$
- $V(X)=n\frac{M}{N}\left(1-\frac{M}{N}\right)\left(\frac{N-n}{N-1}\right)$

Poisson Random Variable $X\sim Poisson(\lambda)$ X denotes the number of events occurring in a fixed period of time or fixed region, λ is the expectation.
- pmf: $f_X(k)=P(X=k)=\frac{e^{-\lambda}\lambda^k}{k!}$, $k\ge 0$, k is the number of occurrences
- $E(X)=\lambda$
- $V(X)=\lambda$
- If $X\sim Poisson(\lambda_1), Y\sim Poisson(\lambda_2), X+Y\sim Poisson(\lambda_1+\lambda_2), (X|X+Y=n)\sim Bin(n,p=\frac{\lambda_1}{\lambda_1+\lambda_2})$

Poisson Process A continuous time process where we count number of occurrences within some interval of time. The defining properties of a Poisson process with rate parameter α are
- the expected number of occurrences in an interval of length T is αT
- there are no simultaneous occurrences
- the number of occurrences in disjoint time intervals are independent

The number of occurrences in any interval T of a Poisson process follows a $Poisson(\alpha T)$ distribution

Poisson Approximation to Binomial Let $X\sim Bin(n,p)$. Suppose $n\to\infty$ and $p\to 0$ such that $\lambda=np$ remains constant. Then $X\sim Poisson(np)$. The approximation is good when $n\ge 20, p\le 0.05$, or $n\ge 100, np\le 10$

Continuous Distributions

Continuous Uniform Distribution $X\sim U(a,b)$
- pdf: $\frac{1}{b-a}, a\le x\le b$
- $E(X)=\frac{a+b}{2}$
- $V(X)=\frac{(b-a)^2}{12}$
- cdf: $\frac{x-a}{b-a}, a\le x\le b$

German Tanks
- The largest number should be close to N
- $M=\max(X_1,...,X_k)\le N$
- $X_1,...,X_k\sim U(0,N)$
- $P(M\le m)=P(X_1\le m,...,X_k\le m)=P(X_1\le m)\cdot ...\cdot P(X_k\le m)$.
- $P(X_i\le m)=\frac{m}{N}, F_M(m)=P(M\le m)=(\frac{m}{N})^k, 0\le m\le N$
- $f_M(m)=\frac{d}{dm}(\frac{m}{N})^k=\frac{km^{k-1}}{N^k}$
- $E(M)=\frac{k}{N^k}\int_0^N m^k dm=\frac{k}{k+1}\cdot N$

Order Statistics
- Suppose $X\sim U(0,\theta)$
- Max: $E[X_n]=\frac{n}{n+1}\theta, V(X_n)=\frac{n}{(n+1)^2(n+2)}\theta^2$
- Min: $E[X_1]=\frac{1}{n+1}\theta, V(X_1)=\frac{n}{(n+1)^2(n+2)}\theta^2$

Exponential Distribution $X\sim Exp(\lambda)$
- pdf: $f_X(x)=\lambda e^{-\lambda x}, x\ge 0, \lambda>0$
- $E(X)=\frac{1}{\lambda}$
- $V(X)=\frac{1}{\lambda^2}$
- cdf: $F_X(x)=1-e^{-\lambda x}, x\ge 0$
- $P(X>x)=e^{-\lambda x}, x>0$
- Alternatively, $\mu=1/\lambda$, $f_X(x)=\frac{1}{\mu}e^{-x/\mu}$, $E(X)=\mu$, $V(X)=\mu^2$, $F_X(x)=1-e^{-x/\mu}$
- $P(X>s+t|X>s)=P(X>t)$, i.e. the distribution is "memoryless"

Normal Distribution $X\sim N(\mu,\sigma^2)$
- pdf: $f_X(x)=\frac{1}{\sqrt{2\pi}\sigma}e^{-(x-\mu)^2/(2\sigma^2)}$
- $E(X)=\mu$
- $V(X)=\sigma^2$
- The total area under the curve is 1
- Two curves with the same σ^2 have identical shapes, but different centers if they have different μ
- As σ increases, the curve flattens, and vice versa
- $X+Y\sim(\mu_1+\mu_2,\sigma_1^2+\sigma_2^2)$

Standardised Normal Distribution Given $X\sim N(\mu,\sigma^2)$, let $Z=\frac{X-\mu}{\sigma}$.
- $Z\sim N(0,1)$, pdf: $f_Z(z)=\frac{1}{\sqrt{2\pi}}exp(-\frac{z^2}{2})$
- For $P(x_1<X<x_2)$, transform it to z_1,z_2. Then $P(x_1<X<x_2)=P(z_1<Z<z_2)$
- $\phi(.)$ and $\Phi(.)$ denote pdf and cdf of the standard normal
- Then $P(x_1<X<x_2)=\Phi(z_2)-\Phi(z_1)$
- $P(Z\ge 0)=P(Z\le 0)=\Phi(0)=0.5$
- $\Phi(z)=P(Z\le z)=P(Z\ge -z)=1-\Phi(-z)$
- $Z\sim N(0,1)\Rightarrow -Z\sim N(0,1)$
- $Z\sim N(0,1)\Rightarrow \sigma Z+\mu\sim N(\mu,\sigma^2)$

Quantile The αth (upper) quartile, $0\le\alpha\le 1$, of X, is the number x_α such that $P(X\ge x_\alpha)=\alpha$. For Z, the αth (upper) quartile or 100α percentage point is $P(Z\ge z_\alpha)=\alpha$.

Normal Approximation to Binomial Let $X\sim Bin(n,p)$, $E(X)=np$, $V(X)=npq$. As $n\to\infty$, $Z=\frac{X-E(X)}{\sqrt{V(X)}}=\frac{X-np}{\sqrt{np(1-p)}}$ is approximately $\sim N(0,1)$, when $np>5$ and $nq>5$

Continuity Correction Accounts for discreteness when approximating to normal, basically when equal, shift outwards, when not equal, shift inwards
- $P(X=k)\approx P(k-0.5<X<k+0.5)$
- $P(a\le X\le b)\approx P(a-0.5<X<b+0.5)$
- $P(a<X<b)\approx P(a+0.5<X<b-0.5)$

Sampling and Sampling Distributions

Population All possible outcomes or observations

Sample Subset of a population

Simple Random Sample n members chosen such that every subset of n observations of the population has the same probability of being selected

SRS from Infinite Population Let X be a random variable with pdf $f_X(x)$. $(X_1,X_2,...,X_n)$ where X_i is an independent random variable with the same distribution as X is a random sample of size n. The joint probability function is $f_{X_1,X_2,...,X_n}(x_1,x_2,...,x_n)=f_{X_1}(x_1)f_{X_2}(x_2)...f_{X_n}(x_n)$. Sampling from a finite population with replacement is the same as sampling from an infinite population if each element has the same probability of being selected and successive draws are independent.

Statistic A function of a random sample of n observations

Sample Mean $\overline{X}=\frac{1}{n}\sum_{i=1}^n X_i$

Sample Variance $S^2=\frac{1}{n-1}\sum_{i=1}^n(X_i-\overline{X})^2$

Sampling Distribution Probability distribution of a statistic

Mean and Variance of $\overline{X}$ For random samples of size n taken from an infinite population with mean μ_X and variance σ_X^2, the sampling distribution of the sample mean $\overline{X}$ has mean μ_X and variance $\frac{\sigma_X^2}{n}$, or $\mu_{\overline{X}}=E(\overline{X})=\mu_X$ and $\sigma_{\overline{X}}^2=V(\overline{X})=\frac{\sigma_X^2}{n}$. $\overline{X}$ can serve as an estimator and as n increases it becomes a more accurate estimator.

Standard Error Spread of a sampling distribution described by its standard deviation. The standard error of $\overline{X}=\sigma_{\overline{X}}=\frac{\sigma_X}{\sqrt{n}}$, describing how much $\overline{x}$ varies between samples.

Law of Large Numbers For independent $X_1,X_2,...,X_n$ with mean μ and variance σ^2, for any $\varepsilon\in\mathbb{R}$, $P(|\overline{X}-\mu|>\varepsilon)\to 0$ as $n\to\infty$

Central Limit Theorem If $\overline{X}$ is the mean of a random sample of size n taken from a population with mean μ and finite variance σ^2, then as $n\to\infty$, $\frac{\overline{X}-\mu}{\sigma/\sqrt{n}}\to Z\sim N(0,1)$, $\overline{X}\to N(\mu,\frac{\sigma^2}{n})$

χ^2 **Distribution** A random variable with the same distribution as Z^2 is a χ^2 random variable with one degree of freedom. A random variable with the same distribution as $Z_1^2+...+Z_n^2$ is a χ^2 random variable with n degrees of freedom, $\chi^2(n)$.
- If $Y\sim\chi^2(n)$, $E(Y)=n$, $V(Y)=2n$
- For large n, $\chi^2(n)\sim N(n,2n)$
- For independent Y_1,Y_2, $Y_1+Y_2\sim\chi^2(m+n)$
- χ^2 is a family of curves with varying degrees of freedom, with density functions having a long right tail
- Define $\chi^2(n;\alpha)$ such that for $Y\sim\chi^2(n)$, $P(Y>\chi^2(n;\alpha))=\alpha$

Sampling Distribution If S^2 is the variance of a random sample of size n taken from a normal population with variance σ^2, $\frac{(n-1)S^2}{\sigma^2}=\frac{\sum_{i=1}^n(X_i-\overline{X})^2}{\sigma^2}\sim\chi^2(n-1)$

t**-Distribution** Suppose $Z\sim N(0,1)$ and $U\sim\chi^2(n)$. If they are independent, then $T=\frac{Z}{\sqrt{U/n}}$, a t-distribution with n degrees of freedom, $t(n)$
- If $X\sim N(\mu,\sigma^2)$, then $T=\frac{\bar{X}-\mu}{S/\sqrt{n}}\sim t_{n-1}$
- As $n\to\infty$, $t(n)\to N(0,1)$ $(n\ge 30)$
- If $T\sim t(n)$, $E(T)=0$, $V(T)=n/(n-2)$, $n>2$
- The graph is symmetric about the vertical axis and resembles Z
- Define $t_{n;\alpha}$ such that for $T\sim t(n)$, $P(T>t_{n;\alpha})=\alpha$

Sampling Distribution For independent and identically distributed normal variables with mean μ and variance σ^2, then $\frac{\overline{X}-\mu}{S/\sqrt{n}}\sim t(n-1)$

F**-Distribution** Suppose $U\sim\chi^2(m)$ and $V\sim\chi^2(n)$ are independent. Then $F=\frac{U/m}{V/n}$, a F-distribution with (m,n) degrees of freedom, $F(m,n)$
- If $X\sim F(m,n)$, then $E(X)=n/(n-2)$, $n>2$, and $V(X)=\frac{2n^2(m+n-2)}{m(n-2)^2(n-4)}$, $n>4$
- If $F\sim F(n,m)$, $1/F\sim F(m,n)$
- As $m\to\infty$, $\frac{U}{m}\to 1$
- $F(m,n;\alpha)$ such that $P(F>F(m,n;\alpha))=\alpha$ where $F\sim F(m,n)$
- $F(m,n;1-\alpha)=1/F(n,m;\alpha)$
- $F=\frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2}\sim F(n_1-1,n_2-1)$
- $T^2\sim F(1,n)$

Estimation

Estimator Rule telling us how to calculate an estimate based on sample information

Unbiased Estimator Let $\hat{\Theta}$ be an estimator of θ. Then $\hat{\Theta}$ is a random varibale based on the sample. If $E(\hat{\Theta}) = \theta$, $\hat{\Theta}$ is an unbiased estimator of θ

Max and Min estimator $\frac{n+1}{n}M, (n+1)m$ are unbiased estimators where $M = \max, m = \min$

Upper-Tail Probability z_α represents $P(Z > z_\alpha) = \alpha$

Maximum Error of Estimate With probability $1-\alpha$, $|\overline{X} - \mu| < E = z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$

Sample Size For maximum error E_0, $n \geq (\frac{z_{\alpha/2}\cdot\sigma}{E_0})^2$

Sampling Cases

Different Cases

	Population	σ	n	Statistic	E	n for desired E_0 and α
I	Normal	known	any	$Z = \frac{\overline{X}-\mu}{\sigma/\sqrt{n}}$	$z_{\alpha/2}\cdot\frac{\sigma}{\sqrt{n}}$	$\left(\frac{z_{\alpha/2}\cdot\sigma}{E_0}\right)^2$
II	any	known	large	$Z = \frac{\overline{X}-\mu}{\sigma/\sqrt{n}}$	$z_{\alpha/2}\cdot\frac{\sigma}{\sqrt{n}}$	$\left(\frac{z_{\alpha/2}\cdot\sigma}{E_0}\right)^2$
III	Normal	unknown	small	$T = \frac{\overline{X}-\mu}{S/\sqrt{n}}$	$t_{n-1;\alpha/2}\cdot\frac{s}{\sqrt{n}}$	$\left(\frac{t_{n-1;\alpha/2}\cdot s}{E_0}\right)^2$
IV	any	unknown	large	$Z = \frac{\overline{X}-\mu}{S/\sqrt{n}}$	$z_{\alpha/2}\cdot\frac{s}{\sqrt{n}}$	$\left(\frac{z_{\alpha/2}\cdot s}{E_0}\right)^2$

Confidence Interval For confidence level $(1-\alpha)$, $P(a < \mu < b) = 1-\alpha$, and (a, b) is the confidence interval

Confidence Intervals for the Mean

The table below gives the $(1-\alpha)$ confidence interval (formulas) for the population mean.

Case	Population	σ	n	Confidence Interval
I	Normal	known	any	$\overline{x} \pm z_{\alpha/2}\cdot\sigma/\sqrt{n}$
II	any	known	large	$\overline{x} \pm z_{\alpha/2}\cdot\sigma/\sqrt{n}$
III	Normal	unknown	small	$\overline{x} \pm t_{n-1;\alpha/2}\cdot s/\sqrt{n}$
IV	any	unknown	large	$\overline{x} \pm z_{\alpha/2}\cdot s/\sqrt{n}$

Note that n is considered large when $n \geq 30$.

Independent Samples Complete randomisation

Matched Pair Samples Randomisation between matched pairs

Independent Samples: Known, Unequal Variances

1. Random sample of size n_1/n_2 from population $1/2$ with mean μ_1/μ_2 and variance σ_1^2/σ_2^2
2. The samples are independent
3. The population variances are known and unequal
4. Either both populations are normal or both samples are large
5. $E(\overline{X} - \overline{Y}) = \mu_1 - \mu_2 = \delta$
6. $V(\overline{X} - \overline{Y}) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$
7. Then $Z = \frac{(\overline{X}-\overline{Y})-(\mu_1-\mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1}+\frac{\sigma_2^2}{n_2}}}$
8. The $100(1-\alpha)\%$ confidence interval for $\mu_1 - \mu_2$ is $(\overline{x}-\overline{y}) \pm z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$

Independent Samples: Large, Unknown Variances

1. Random sample of size n_1/n_2 from population $1/2$ with mean μ_1/μ_2 and variance σ_1^2/σ_2^2
2. The samples are independent
3. The population variances are unknown and unequal
4. Both samples are large
5. The $100(1-\alpha)\%$ confidence interval for $\mu_1 - \mu_2$ is $(\overline{X}-\overline{Y}) \pm z_{\alpha/2}\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}$

Independent Samples: Small, Equal Variances

1. Random sample of size n_1/n_2 from population $1/2$ with mean μ_1/μ_2 and variance σ_1^2/σ_2^2
2. The samples are independent
3. The population variances are unknown and the same
4. Both samples are small
5. Both populations are normally distributed
6. σ^2 can be estimated by the pooled sample variance $S_p^2 = \frac{(n_1-1)S_1^2+(n_2-1)S_2^2}{n_1+n_2-2}$
7. $T = \frac{(\overline{X}-\overline{Y})-(\mu_1-\mu_2)}{S_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}} \sim t_{n_1+n_2-2}$
8. The $100(1-\alpha)\%$ confidence interval for $\mu_1 - \mu_2$ is $(\overline{X}-\overline{Y}) \pm t_{n_1+n_2-2;\alpha/2}S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$

Independent Samples: Large, Equal Variances

1. Replace $t_{n_1+n_2-2;\alpha/2}$ with $z_{\alpha/2}$ in the previous formula
2. The $100(1-\alpha)\%$ confidence interval for $\mu_1 - \mu_2$ is $(\overline{X}-\overline{Y}) \pm z_{\alpha/2}S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$
3. Equal Variance can be assumed if $0.5 \leq S_1/S_2 \leq 2$

Paired Data

1. $(X_1, Y_1), ..., (X_n, Y_n)$ are matched pairs where X_i/Y_i is a random sample from population $1/2$
2. X_i and Y_i are dependent
3. (X_i, Y_i) and (X_j, Y_j) are independent for $i \neq j$
4. For matched pairs define $D_i = X_i - Y_i, \mu_D = \mu_1 - \mu_2$
5. We can treat $D_1, ..., D_n$ as a random sample from a single population with mean μ_D and variance σ_D^2
6. Consider $T = \frac{\overline{D}-\mu_D}{S_D/\sqrt{n}}$
7. For small samples, if the population is normally distributed then $T \sim t_{n-1}$ and the confidence interval for μ_D is $\overline{d} \pm t_{n-1;\alpha/2}\cdot\frac{s_D}{\sqrt{n}}$
8. For large samples then $T \sim N(0,1)$ and the confidence interval for μ_D is $\overline{d} \pm z_{\alpha/2}\cdot\frac{s_D}{\sqrt{n}}$

Hypothesis Tests

How?

1. Set null and alternative hypotheses
2. Set level of significance
3. Identify test statistic, distribution, and rejection criteria
4. Compute observed test statistic value
5. Conclude

Null v/s Alternative Hypotheses We adopt the position of the null hypothesis unless there is overwhelming evidence against it. The hypothesis to be proven is the alternative hypothesis which often states the null hypothesis is false.

Type I/II Errors Type I: Rejecting H_0 when H_0 is true, Type II: Not rejecting H_0 when H_0 is false.

Level of Significance / Power Level of Significance: $\alpha = P(\text{Type I Error})$, Power: $1-\beta = P(\text{Reject } H_0 \mid H_0 \text{ is False})$ where $\beta = P(\text{Type II Error})$

Hypothesis Test for Mean: Known Variance

1. The population variance σ^2 is known and the underlying distribution is normal or sufficiently large
2. For $H_0 : \mu = \mu_0$, $Z = \frac{\overline{X}-\mu_0}{\sigma/\sqrt{n}} \sim N(0,1)$
3. Let z be the observed Z value, then for the alternative hypothesis
4. $H_1 : \mu \neq \mu_0$, rejection region: $z < -z_{\alpha/2}$ or $z > z_{\alpha/2}$
5. $H_1 : \mu < \mu_0$, rejection region: $z < -z_\alpha$
6. $H_1 : \mu > \mu_0$, rejection region: $z > z_\alpha$

p-value

- Probability of obtaining a test statistic at least as extreme as the observed sample value given H_0 is true, a.k.a observed level of significance
- For 2 sided: $P(|Z| > |z|) = 2P(Z > |z|) = 2P(Z < -|z|)$
- 1 sided: $P(Z < -|z|)$ or $P(Z > |z|)$
- If p-value $< \alpha$, reject H_0, if not do not reject H_0

Hypothesis test for Mean: Unknown Variance

1. The population variance σ^2 is unknown and the underlying distribution is normal
2. For $H_0 : \mu = \mu_0$, $T = \frac{\overline{X}-\mu_0}{S/\sqrt{n}} \sim t_{n-1}$
3. Let t be the observed T value, then for the alternative hypothesis
4. $H_1 : \mu \neq \mu_0$, rejection region: $t < -t_{n-1;\alpha/2}$ or $t > t_{n-1;\alpha/2}$
5. $H_1 : \mu < \mu_0$, rejection region: $t < -t_{n-1;\alpha}$
6. $H_1 : \mu > \mu_0$, rejection region: $t > t_{n-1;\alpha}$
7. When n is large, t_{n-1} can be replaced by Z

Comparing Means: Independent Samples I

1. The population variances σ_1^2, σ_2^2 are known and the underlying distributions are normal or samples are sufficiently large
2. For $H_0 : \mu_1 - \mu_2 = \delta_0$, $Z = \frac{(\overline{X}-\overline{Y})-\delta_0}{\sqrt{\frac{\sigma_1^2}{n_1}+\frac{\sigma_2^2}{n_2}}} \sim N(0,1)$
3. When population variances are unknown and samples are sufficiently large
4. For $H_0 : \mu_1 - \mu_2 = \delta_0$, $Z = \frac{(\overline{X}-\overline{Y})-\delta_0}{\sqrt{\frac{S_1^2}{n_1}+\frac{S_2^2}{n_2}}} \sim N(0,1)$

Rejection Regions and p-Values

For the null hypothesis $H_0 : \mu_1 - \mu_2 = \delta_0$, and specified alternative H_1, the rejection regions and p-values are given below.

H_1	Rejection Region	p-value
$\mu_1 - \mu_2 > \delta_0$	$z > z_\alpha$	$P(Z > \lvert z\rvert)$
$\mu_1 - \mu_2 < \delta_0$	$z < -z_\alpha$	$P(Z < -\lvert z\rvert)$
$\mu_1 - \mu_2 \neq \delta_0$	$z > z_{\alpha/2}$ or $z < -z_{\alpha/2}$	$2P(Z > \lvert z\rvert)$

Comparing Means: Independent Samples II

1. The population variances σ_1^2, σ_2^2 are unknown but equal and the underlying distributions are normal and the samples are small
2. For $H_0 : \mu_1 - \mu_2 = \delta_0$, $Z = \frac{(\overline{X}-\overline{Y})-\delta_0}{S_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}} \sim t_{n_1+n_2-2}$

Alt (H_1)	Reject H_0 if...	p-value
$\mu_1 - \mu_2 > \delta_0$	$T > t_{\alpha,\nu}$	$P(T_{obs} > T)$
$\mu_1 - \mu_2 < \delta_0$	$T < -t_{\alpha,\nu}$	$P(T_{obs} < T)$
$\mu_1 - \mu_2 \neq \delta_0$	$T > t_{\alpha/2,\nu}$ or $T < -t_{\alpha/2,\nu}$	$2P(T_{obs} > \|T\|)$

Comparing Means: Paired Data

1. For paired data, $D_i = X_i - Y_i$
2. For $H_0 : \mu_D = \mu_{D_0}, T = \frac{\overline{D} - \mu_{D_0}}{S_D/\sqrt{n}}$
3. If the sample is small and population is normally distributed then $T \sim t_{n-1}$
4. If the sample is large then $T \sim N(0, 1)$

Alt (H_1)	Reject H_0 if...	p-value
$\mu_D > \mu_{D_0}$	$T > t_{\alpha,n-1}$	$P(T_{obs} > T)$
$\mu_D < \mu_{D_0}$	$T < -t_{\alpha,n-1}$	$P(T_{obs} < T)$
$\mu_D \neq \mu_{D_0}$	$T > t_{\alpha/2,n-1}$ or $T < -t_{\alpha/2,n-1}$	$2 \times P(T_{obs} > \|T\|)$